

# YANQI SU

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## RESEARCH INTERESTS

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My research interests include **Software Testing**, **Program Analysis**, **Knowledge Graphs** and **Large Language Models (LLMs)** for Software Engineering, and **Software Repository Mining**.

## EDUCATION

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| <b>The Australian National University (ANU)</b><br>Ph.D. in Computer Science<br>Supervisor: Zhenchang Xing | <i>Sep. 2020 - Sep. 2024</i> |
| <b>Nanjing University of Aeronautics and Astronautics (NUAA)</b><br>M.E. in Computer Science               | <i>Sep. 2017 - Jun. 2020</i> |
| <b>Nanjing University of Aeronautics and Astronautics (NUAA)</b><br>B.E. in Computer Science               | <i>Sep. 2013 - Jun. 2017</i> |

## PUBLICATIONS

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- Yanqi Su**, Michael Pradel, Chunyang Chen: RippleGUItester: Change-Aware Exploratory Testing. The 35th SIGSOFT International Symposium on Software Testing and Analysis (ISSTA 2026). Acceptance rate: 10.1% (90/888)
- Yanqi Su**, Zhenchang Xing, Chong Wang, Chunyang Chen, Xiwei Xu, Qinghua Lu, Liming Zhu: Automated Soap Opera Testing Directed by LLMs and Scenario Knowledge: Feasibility, Challenges, and Road Ahead. The ACM International Conference on the Foundations of Software Engineering (FSE2025).
- Yanqi Su**, Dianshu Liao, Zhenchang Xing, Qing Huang, Mulong Xie, Qinghua Lu, Xiwei Xu: Enhancing Exploratory Testing by Large Language Model and Knowledge Graph. The 46th International Conference on Software Engineering (ICSE'24).
- Yanqi Su**, Zheming Han, Zhenchang Xing, Xiwei Xu, Liming Zhu, Qinghua Lu: SoapOperaTG: A Tool for System Knowledge Graph Based Soap Opera Test Generation. The 45th International Conference on Software Engineering (ICSE'23 Tool Demo).
- Yanqi Su**, Zheming Han, Zhenchang Xing, Xin Xia, Xiwei Xu, Liming Zhu, Qinghua Lu: Constructing a System Knowledge Graph of User Tasks and Failures from Bug Reports to Support Soap Opera Testing. 37th IEEE/ACM International Conference on Automated Software Engineering (ASE2022).
- Yanqi Su**, Zhenchang Xing, Xin Peng, Xin Xia, Chong Wang, Xiwei Xu, Liming Zhu: Reducing Bug Triaging Confusion by Learning from Mistakes with a Bug Tossing Knowledge Graph. 36th IEEE/ACM International Conference on Automated Software Engineering (ASE2021).
- 🏆 **Distinguished Paper Award**
- Yanqi Su**, Zheming Han, Zhipeng Gao, Zhenchang Xing, Qinghua Lu, Xiwei Xu: Still Confusing for Bug-Component Triaging? Deep Feature Learning and Ensemble Setting to Rescue. The 31st IEEE/ACM International Conference on Program Comprehension (ICPC2023).

Yu Zhou, **Yanqi Su**, Taolue Chen, Zhiqiu Huang, Harald C. Gall, Sebastiano Panichella: User Review-Based Change File Localization for Mobile Applications. IEEE Trans. on Software Engineering. (TSE)

Zhipeng Gao, **Yanqi Su**, Xing Hu, Xin Xia: Automating TODO-dropped Methods Detection and Patching. ACM Transactions on Software Engineering and Methodology. (TOSEM)

## TEACHING AND WORKING

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**Postdoctoral Researcher in Technische Universität München (TUM)**, Software Engineering & AI, from Oct-21-2024, supervised by Chunyang Chen

**Lecturer in TUM**, Seminar Exploratory Software Testing (INHN0015, INHN4016, IN0014, IN45111), 2026 Summer

**Lecturer in TUM**, Seminar Exploratory Software Testing (IN0014) (INHN0015, INHN4012), 2025 Summer

**Guest Lecturer in TUM**, Advanced Topics in Software Engineering (CITHN3003), 2024 & 2025 Winter

**Teaching Assistant in ANU**, COMP4130 Managing Software Quality and Process, 2024S1, supervised by Xiaoyu Sun

**Research Visit in University of Stuttgart & CISPA**, Software Lab, from Nov-01-2025 to Nov-12-2025, supervised by Michael Pradel

**Research Assistant in Fudan University**, CodeWisdom Team, from Sep-01-2020 to Jun-30-2021, supervised by Xin Peng

**Teaching Assistant in NUAA**, Software Metrics, 2017S2, supervised by Yu Zhou

## MENTOR

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**2025 Abdelrahman Diab, TUM** Master thesis: Bug-Inducing Code Change Detection

**2025 Lazar Minic, TUM** Master thesis: Bridging Policy and Practice: Leveraging LLMs for Privacy-Aware Exploratory Software Testing

**2024 Alper Temel, TUM** Master thesis: Non-intrusive GUI Element Location Using LLMs

**2024 Linsen Mao, TUM** Bachelor thesis: Soap Opera Testing on Enterprise Resource Planning Software

**2024 Bryan Alvin, TUM** Bachelor thesis: Automated Exploratory Testing of Business Logic in ERP Systems

**2022 Zheming Han, ANU** Master thesis: Reducing Bug Component Triaging Mistakes by Deep Learning

**2017 Jian Liu, NUAA** Bachelor thesis: Code Reviewer Recommendation Based on File Path Analysis

## FUNDINGS & AWARDS

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ANU Vice-Chancellor's Travel Grant

ACM SIGSOFT CAPS AWARD FSE/ISSTA 2025

ACM Travel Grant for 2024 IEEE/ACM 46th International Conference on Software Engineering

ACM SIGSOFT Distinguished Paper Award (First author) in ASE2021

HDR Fee Remission Merit Scholarship, ANU, 2020.09 – 2024.09

University Research Scholarship, ANU, 2020.09 – 2024.09

First-Class Graduate Scholarship, NUA, 2017.09 – 2020.04

Outstanding Research and Innovation Award, NUA, 2019.09 – 2020.04

Outstanding Contribution to Social Activities, NUA, 2019.09 – 2020.04

Third-Class Scholarship, NUA, 2015.09 – 2017.06

Second-Class Scholarship, NUA, 2014.09 – 2015.06

Third-Class Scholarship, NUA, 2013.09 – 2014.06

Microsoft Research Asia Club Star, 2014

Outstanding Youth Volunteer, 2nd Youth Olympic Games, 2014.08 – 2014.09

## SERVICE

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### Journal Reviewer

- IEEE Transactions on Software Engineering (TSE)
- ACM Transactions on Software Engineering and Methodology (TOSEM)
- Empirical Software Engineering (EMSE)
- Software Testing, Verification and Reliability (STVR)

### Conference Program Committee

- The ACM International Conference on the Foundations of Software Engineering (FSE2027)
- The 49th IEEE/ACM International Conference on Software Engineering (ICSE2027)
- The 41th IEEE/ACM International Conference on Automated Software Engineering (ASE2026)
- The 23rd International Conference on Mining Software Repositories (MSR2026)
- The 17th International Conference on Internetware (Internetware 2026)
- The International Conference on Predictive Models and Data Analytics in Software Engineering (PROMISE 2026)
- The 1st International Workshop on User Interface and Experience for Software Engineering (UISE 2026)
- The 40th IEEE/ACM International Conference on Automated Software Engineering (ASE2025)

- The 36th IEEE International Symposium on Software Reliability Engineering (ISSRE2025)
- The 32nd Asia-Pacific Software Engineering Conference (APSEC2025 SEIP Track & Student Research Competition Track)

### **Student Volunteer**

- The ACM International Conference on the Foundations of Software Engineering (FSE2025)

## **TALK**

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### **Shaping the Future of Exploratory Software Testing with Knowledge Graphs and LLMs**

This talk is a compilation of my main research and I gave this talk at multiple places.

- Jul. 2025 The Chinese University of Hong Kong, Shenzhen, China
- Jun. 2025 University of Oxford, UK
- Apr. 2025 Aalto University, Finland

### **Automated Soap Opera Testing Directed by LLMs and Scenario Knowledge**

- Feb. 2025 Technical University of Munich, Germany

### **Constructing a System Knowledge Graph of User Tasks and Failures to Support Soap Opera Testing**

- Dec. 2023 Google Research Australia, Australia

## **REFERENCES**

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### **Zhenchang Xing** (Supervisor)

Associate Professor  
School of Computing  
The Australian National University  
Senior Principal Research Scientist  
CSIRO's Data61  
Australia  
Zhenchang.Xing@data61.csiro.au

### **Xin Peng**

Deputy Dean & Professor  
School of Computer Science  
Fudan University, China  
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### **Chunyang Chen**

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